ENGINEERING ECONOMY

LECTURE 1

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Engineering Economy:

Engineering Economy is A collection of mathematical techniques that simplify economic comparison and assist people in making decisions.

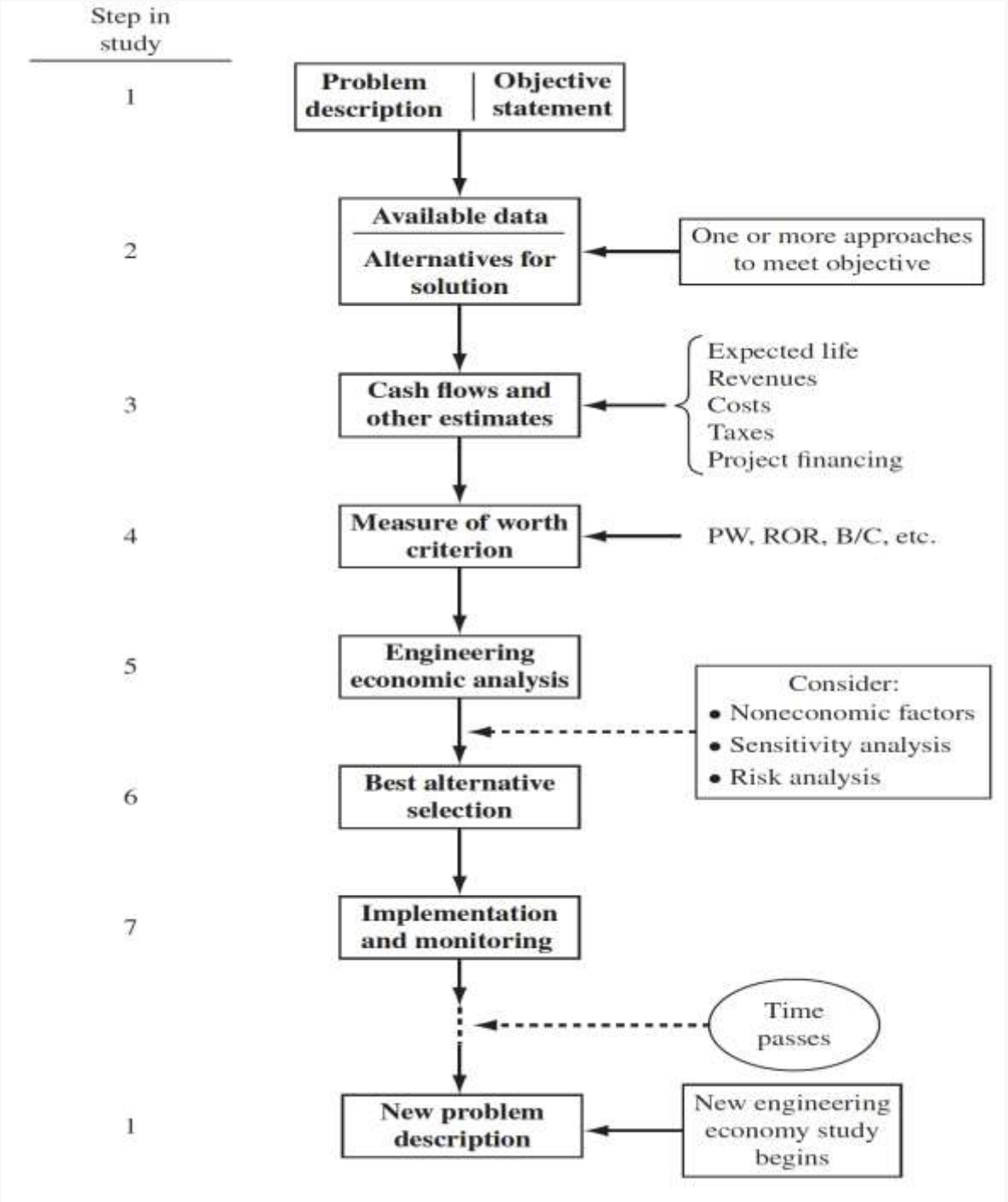
Fundamentally, engineering economy involves formulating, estimating and evaluating the economic outcomes when alternatives to accomplish a defined purpose are available.

The need for engineering economy is primarily motivated by the work that engineers do in performing analyses, synthesizing, and coming to a conclusion as they work on projects of all sizes.

In other words, engineering economy is at the heart of making decisions. These decisions involve the fundamental elements of **cash flows of money**, **time**, and **interest rates**.

Steps in an Engineering Economy Study

- The steps in an engineering economy study are as follows:
  1. Identify and understand the problem; identify the objective of the project.
  2. Collect relevant, available data and define viable solution alternatives.
  3. Make realistic cash flow estimates.
  4. Identify an economic measure of worth criterion for decision making.
  5. Evaluate each alternative; consider noneconomic factors; use sensitivity analysis as needed.
  6. Select the best alternative.
  7. Implement the solution and monitor the results.


Step in study
1
Problem description
Objective statement
2
Available data
Alternatives for solution
One or more approaches to meet objective
3
Cash flows and other estimates
Expected life
Revenues
Costs
Taxes
Project financing
4
Measure of worth criterion
PW, ROR, B/C, etc.
5
Engineering economic analysis
Consider:
• Noneconomic factors
• Sensitivity analysis
• Risk analysis
6
Best alternative selection
7
Implementation and monitoring
Time passes
1
New problem description
New engineering economy study begins

- Descriptions of several of the elements in the steps are important to understand.
- Problem Description and Objective Statement: A clear, short description of the problem and main goals is essential for finding other possible solutions.
- Ex:
- Problem Description: A coal-fueled power plant must be shut down by 2025 due to the production of excessive sulfur dioxide.
- Objectives: Generate the forecasted electricity needed for 2025 and beyond, plus to not exceed all the projected emission allowances in these future years.

- Alternatives: These are stand-alone descriptions of viable solutions to problems that can meet the objectives. Words, pictures, graphs, equipment and service descriptions, simulations, etc. define each alternative. The best estimates for parameters are also part of the alternative.

Some parameters include

- equipment first cost,
- expected life,
- salvage value (estimated trade-in, resale, or market value), and
- annual operating cost (AOC), which can also be termed maintenance and operating (M&O) cost, and
- subcontract cost for specific services. If changes in income (revenue) may occur, this parameter must be estimated.
- Detailing all viable alternatives at this stage is crucial.

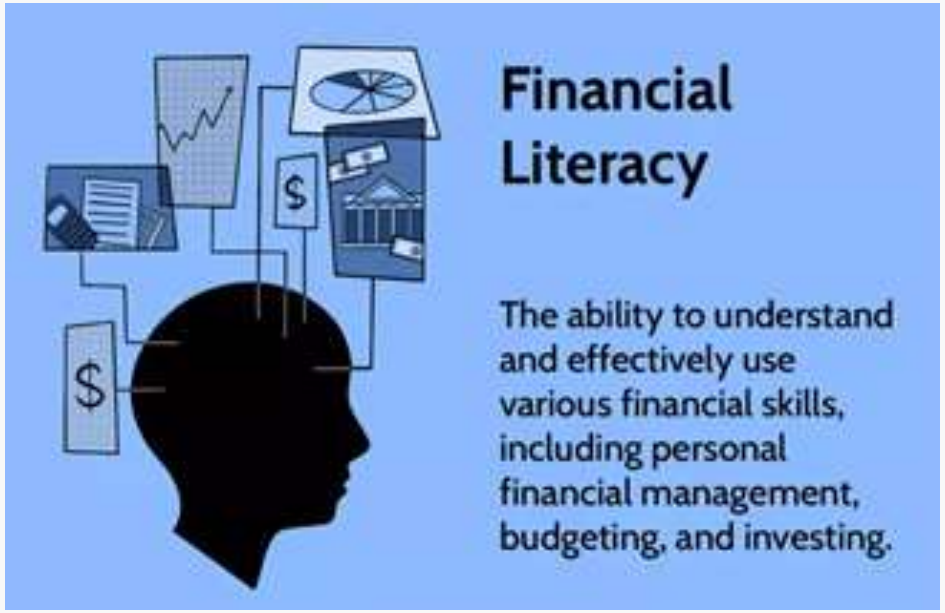
- Cash Flows: All cash flows are estimated for each alternative.
- Since these are future expenditures and revenues, the results of step 3 usually prove to be inaccurate when an alternative is actually in place and operating.
- When cash flow estimates for specific parameters are expected to vary significantly from a point estimate made now, risk and sensitivity analyses (step 5) are needed to improve the chances of selecting the best alternative.
- Sizable variation is usually expected in estimates of revenues, AOC, salvage values, and subcontractor costs.

- Selection of the Best Alternative: The measure of worth is a primary basis for selecting the best economic alternative. For example, if alternative A has a rate of return (ROR) of 15.2% per year and alternative B will result in an ROR of 16.9% per year, B is better economically.
- However, there can always be noneconomic or intangible factors that must be considered and that may alter the decision. There are many possible noneconomic factors; some typical ones are:
  - Market pressures, such as need for an increased international presence
  - Availability of certain resources, e.g., skilled labor force, water, power, tax incentives
  - Government laws that dictate safety, environmental, legal, or other aspects
  - Corporate management's or the board of director's interest in a particular alternative
  - Goodwill offered by an alternative toward a group, for example, employees, union, county, etc.

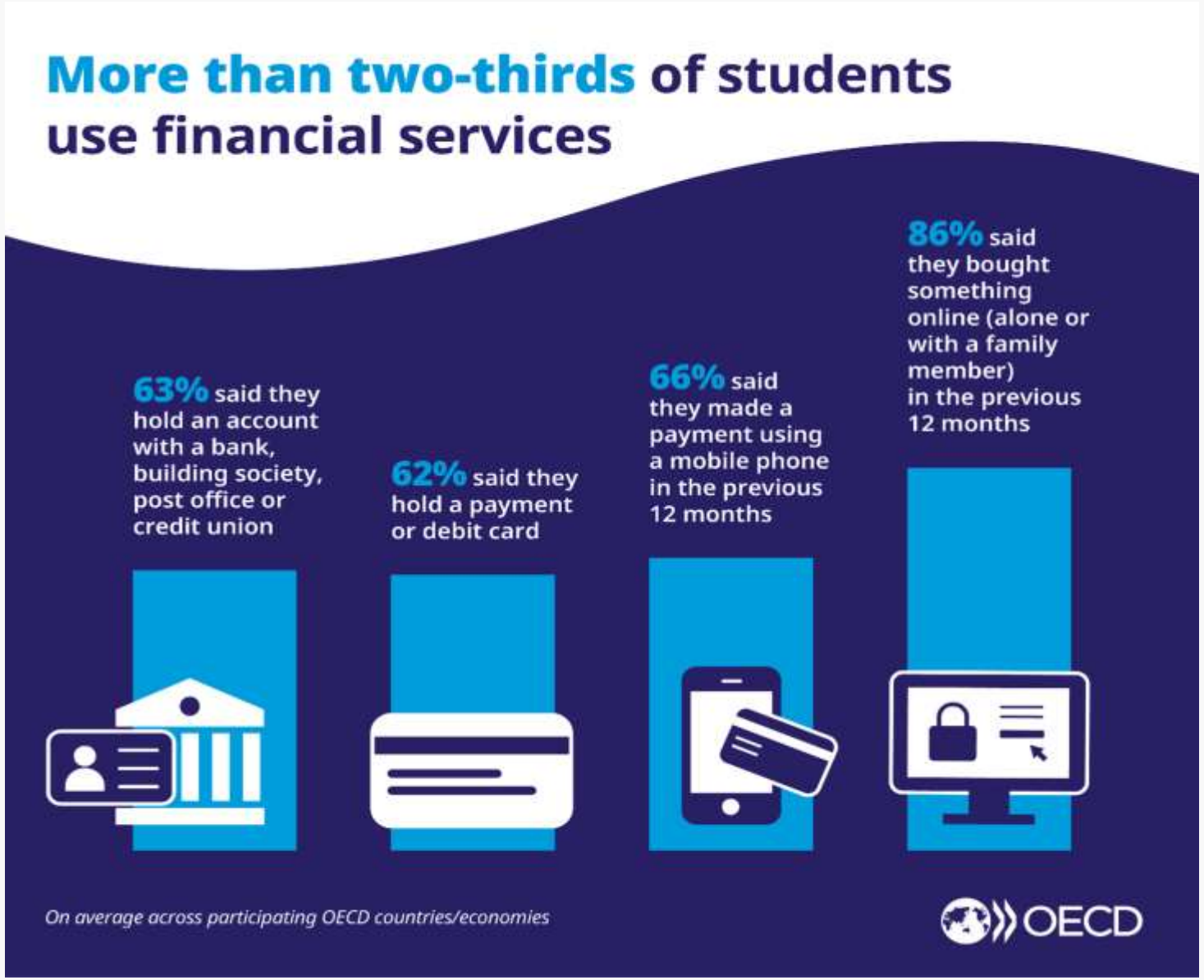
- Whether we are aware of it or not, we use criteria every day to choose between alternatives.
- For example, when you drive to campus or work, you decide to take the “best” route.
- But how did you define best?
- Was the best route the safest, shortest, fastest, cheapest, most scenic, or what?
- Obviously, depending upon which criterion or combination of criteria is used to identify the best, a different route might be selected each time.
- In economic analysis, money (such as dollars or other currency) is usually used as the common measure for evaluating alternatives.
- Thus, when there are several ways of accomplishing a stated objective, the alternative with the lowest overall cost or highest overall net income is selected.

Financial literacy

- To be financially literate is very important as an engineer and as a person.
- Unfortunately many people do not have the fundamental understanding of concepts such as financial risk and diversification, inflation, numeracy, and compound interest.
- **Financial literacy** is the set of skills needed to handle money wisely, invest effectively, and plan for a comfortable future.



- It is the ability to understand and effectively use various financial skills, including personal financial management, budgeting, and investing.



- As our latest PISA report on financial literacy shows,
- more than two thirds of 15-year-olds are active users of financial products and services.
- Over 60% have a bank account and/or a payment card, and
- close to 90% have bought something online during the last 12 months.

- However, their levels of financial literacy remain low: one-fifth of students, on average, do not have basic proficiency in financial literacy, meaning that they are not able to apply their knowledge to real life situations involving financial issues and decisions.
- The fact is many young people simply do not have the knowledge and skills needed to safely navigate the complexities of digital finance.
- Policy makers recognize that young people need to be financially literate to perform common tasks in their day-today lives, such as using a payment card, choosing amongst mobile phone plans or understanding the implications of in-app purchases.
- Students also need basic financial skills to make important decisions about their future, such as understanding different options for funding their education or choosing between different career and education paths.
- As students become increasingly independent from their families, it is likely that the number and complexity of financial decisions will also increase

Comparing countries' and economies' mean performance in financial literacy

Mean score	Comparison country/economy	Countries and economies whose mean score is not statistically significantly different from the comparison country's/economy's score
527	*Flemish community of Belgium*	Denmark*, Netherlands*
521	Denmark*	*Flemish community of Belgium*, *Canadian provinces**, Netherlands*
519	*Canadian provinces**	Denmark*, Netherlands*
517	Netherlands*	*Flemish community of Belgium*, Denmark*, *Canadian provinces**, United States*
507	Czechia	Austria, Poland, United States*
506	Austria	Czechia, Poland, United States*
506	Poland	Czechia, Austria, United States*
505	United States*	Netherlands*, Czechia, Austria, Poland
494	Portugal	Hungary, Norway
492	Hungary	Portugal, Norway, Spain
489	Norway	Portugal, Hungary, Spain, Italy
486	Spain	Hungary, Norway, Italy
484	Italy	Norway, Spain
441	United Arab Emirates	
426	Bulgaria	Peru, Costa Rica
421	Peru	Bulgaria, Costa Rica, Brazil
418	Costa Rica	Bulgaria, Peru, Brazil, Saudi Arabia
416	Brazil	Peru, Costa Rica, Saudi Arabia
412	Saudi Arabia	Costa Rica, Brazil, Malaysia
406	Malaysia	Saudi Arabia

Statistically significantly above the OECD average
Not statistically significantly different from the OECD average
Statistically significantly below the OECD average
Source: OECD, PISA 2022 Database.

Benefits of Financial Literacy

- Broadly speaking, the benefit of financial literacy is that it empowers individuals to make smarter decisions about their finances. In addition:
- Financial literacy can prevent devastating financial mistakes: Naive financial decisions can have devastating effects. The payments on a floating-rate loan can rise unexpectedly. An early withdrawal from an IRA will cost you at tax time. Financially literate people don't make these mistakes.
- Financial literacy prepares people for financial emergencies: Losing a job or having a major unexpected expense is never easy, but the blow is cushioned for those who have an emergency fund.
- Financial literacy can help individuals reach their goals: By understanding how to budget and save money, people define their immediate necessities and focus their minds ahead to important financial goals.
- Financial literacy fosters confidence: Imagine having to make a life-changing decision without all the necessary information. With knowledge about finances, individuals can approach major life choices with greater confidence. They'll be more likely to achieve the outcome they desire and less likely to be surprised or hurt by unforeseen outcomes.

What Are Some Popular Personal Budget Rules?

- Two commonly used personal budgeting methods are the 50/20/30 and 70/20/10 rules, and their simplicity is what makes them popular.

- The first entails dividing your after-tax pay into three areas: needs (50%), savings (20%), and wants (30%).

- The 70/20/10 rule follows a similar blueprint, recommending that your after-tax, take-home income be divided into segments devoted to expenses (70%), savings or reducing debt (20%), and investments and charitable donations (10%).

What Are the Principles of Financial Literacy?

- There are five broad principles of financial literacy.
- financial literacy is to teach individuals about:
- earning,
- spending,
- saving,
- borrowing, and
- protecting their money.

The time value of money

Time Value of Money

Cash flows occur over a substantial period of time. How can we treat the same amount of money available at different times?

- It is a well-known fact that money makes money.
- The time value of money explains the change in the amount of money over time for funds that are owned (invested) or owed (borrowed).

- In general the time value of money means that **a sum of money today is worth more than the same sum in the future**, because money today can be invested to earn interest or returns..

- If we decide to invest capital (money) in a project today, we inherently expect to have more money in the future than we invested.
- If we borrow money today, in one form or another, we expect to return the original amount plus some additional amount of money.
- An engineering economic analysis can be performed on future estimated amounts or on past cash flows to determine if a specific measure of worth, e.g., rate of return, was achieved.

EXAMPLE

- Which would you rather have?
- $100 in your hand right now
- A guaranteed $100, but only one year from today

Interest Rate and Rate of Return

- Interest refers to the cost of borrowing money or the reward for lending money.
- In the financial world, money itself is a commodity, and like other goods that are bought and sold, money costs money.
- There are always two perspectives to an amount of interest—interest paid and interest earned.
- Interest is paid when a person or organization borrowed money (obtained a loan) and repays a larger amount over time.
- Interest is earned when a person or organization saved, invested, or lent money and obtains a return of a larger amount over time.

- **Interest** paid on borrowed funds (a loan) is determined using the original amount, also called the principal,

$$\text{Interest} = \text{amount owed now} - \text{principal}$$

- When interest paid over a specific time unit is expressed as a percentage of the principal, the result is called the **interest rate**.

$$\text{Interest rate (\%)} = \frac{\text{interest accrued per time unit}}{\text{principal}} \times 100\%$$

- The time unit of the rate is called the interest period.
- By far the most common interest period used to state an interest rate is 1 year.
- Shorter time periods can be used, such as 1% per month.
- Thus, the interest period of the interest rate should always be included. If only the rate is stated, for example, 8.5%, a 1-year interest period is assumed.

EXAMPLE

- An employee at LaserKinetics.com borrows $10,000 on May 1 and must repay a total of $10,700 exactly 1 year later. Determine the interest amount and the interest rate paid.

Solution

The perspective here is that of the borrower since $10,700 repays a loan. to determine the interest paid.

$$\text{Interest paid} = \$10{,}700 - 10{,}000 = \$700$$

interest rate paid for 1 year.

$$\text{Percent interest rate} = \frac{\$700}{\$10{,}000} \times 100\% = 7\% \text{ per year}$$

EXAMPLE

- Stereophonics, Inc. plans to borrow $20,000 from a bank for 1 year at 9% interest for new recording equipment.

  (a) Compute the interest and the total amount due after 1 year.

  (b) Construct a column graph that shows the original loan amount and total amount due after 1 year used to compute the loan interest rate of 9% per year.

Solution

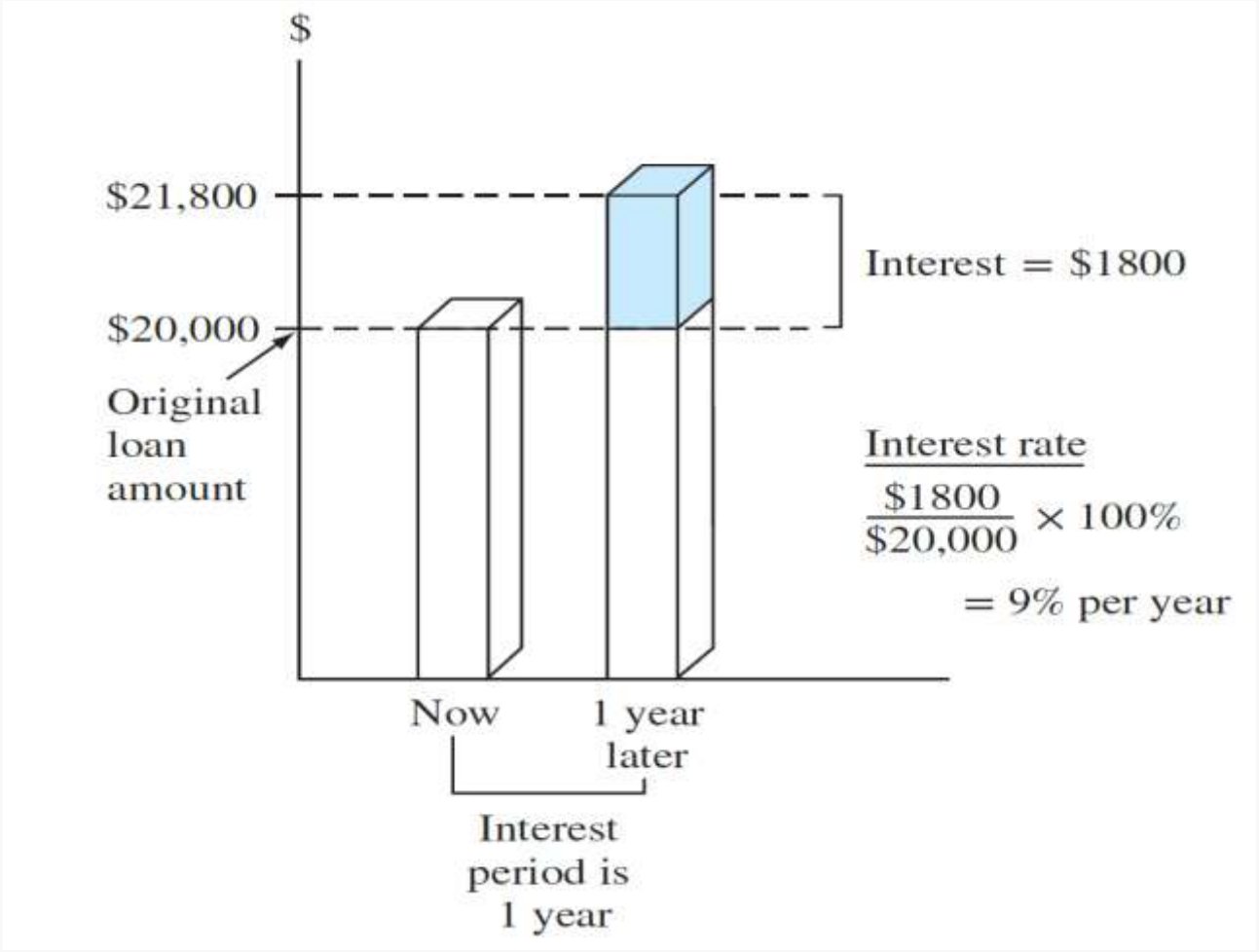
(a) Compute the total interest accrued by solving Equation for interest accrued.

$$\text{Interest} = \$20{,}000(0.09) = \$1800$$

The total amount due is the sum of principal and interest.

$$\text{Total due} = \$20{,}000 + 1800 = \$21{,}800$$

(b) $1800 interest, $20,000 original loan principal, 1-year interest period.



- (Note that in part (a), the total amount due may also be computed as

  Total due = principal(1 + interest rate) = $20,000(1.09) = $21,800

- From the perspective of a saver, a lender, or an investor, interest earned is the final amount minus the initial amount, or principal.

  Interest earned = total amount now − principal

- Interest earned over a specific period of time is expressed as a percentage of the original amount and is called **rate of return (ROR).**

$$\text{Rate of return (\%)} = \frac{\text{interest accrued per time unit}}{\text{principal}} \times 100\%$$

- The time unit for rate of return is called the interest period, just as for the borrower's perspective. Again, the most common period is 1 year.
- The term return on investment (ROI) is used equivalently with ROR in different industries and settings, especially where large capital funds are committed to engineering-oriented programs.
- The term *interest rate paid* is more appropriate for the borrower's perspective, while the term *rate of return earned* applies for the investor's perspective.

EXAMPLE

- (a) Calculate the amount deposited 1 year ago to have $1000 now at an interest rate of 5% per year.
- (b) Calculate the amount of interest earned during this time period.

Solution

(*a*) The total amount accrued ($1000) is the sum of the original deposit and the earned interest.

If X is the original deposit,

$$\text{Total accrued} = \text{deposit} + \text{deposit(interest rate)}$$

$$\$1000 = X + X(0.05) = X(1 + 0.05) = 1.05X$$

The original deposit is

$$X = \frac{1000}{1.05} = \$952.38$$

(*b*) to determine the interest earned.

$$\text{Interest} = \$1000 - 952.38 = \$47.62$$

Simple interest and Compound interest

- In these examples and the interest amount was calculated at the end of one period.
- When more than one interest period is involved, e.g., the amount of interest after 3 years, it is necessary to state whether the interest is accrued on a simple or compound basis from one period to the next.

- **Simple interest** is calculated using the principal only, ignoring any interest accrued in preceding interest periods.

Interest = (Principal)(# of periods)(interest rate)

Example

321 Credit Union loaned money to an engineer. The loan is for $1,000 for 3 years at 5% per year simple interest. How much money will the engineer repay at the end of 3 years?

Solution

- Interest per year = 1000(0.05)=$50
- Total Interest = 1000(3)(0.05) = $150
- The amount due after 3 years = 1000+150=1150
- OR: $P(1+i * n)$= 1000(1+3*0.05)=1000(1.15)

Simple Interest Calculations

End of Year	Amount Borrowed	Interest	Amount Owed	Amount Paid
0	\$1000			
1	-	$1000 * 0.05 = 50$	1050	-
2	-	$1000 * 0.05 = 50$	1100	-
3	-	$1000 * 0.05 = 50$	1150	1150

- For **Compound interest rate**, the interest is accrued on the principal plus the total amount of interest accumulated in all previous periods.

Interest = (principal+all accumulated interest)(interest rate)

- Therefore, compound interest means interest on top of interest (interest earned earns interest)

Example

Same as previous example but the loan is at 5% per year compound interest.

Compound Interest Calculations

End of Year	Amount Borrowed	Interest	Amount Owed	Amount Paid
0	\$1000			
1	-	$1000 * 0.05$=50	1050	-
2	-	$1050 * 0.05 = 52.5$	1102.5	-
3	-	$1102.5 * 0.05 = 55.13$	1157.63	1157.63

Note on Calculating Total Amount Due

- Year 1: $1,000(1.05) = 1,050$
- Year 2: $1,050(1.05) = 1,000(1.05)(1.05) = 1102,5$
- Year 3: $1102,5(1.05) = 1,000(1.05)(1.05)(1.05) = 1157.63$

- Therefore, in general the formula form:

$$\text{Total due} = \text{Principal}\,(1 + \text{interest rate})^{\text{number of years}}$$

In Class Work 1

Calculate the interest and payment amounts for a $5,000 loan which will be repaid in 5 years at 10% interest rate under the following plans:

1. Simple interest, pay all at end
2. Compound interest, pay all at end
3. Simple interest paid annually, principal paid at end
4. Compound interest and portion of principal ($1,000) repaid annually

1) Simple interest, pay all at end

Plan 1

End of Year	Interest for Year	Owed at End of Year	End of Year Payment	Owed after Payment
0				$5,000
1	$5,000 * 0.1 = 500$	5,500	-	5,500
2	$5,000 * 0.1 = 500$	6,000	-	6,000
3	$5,000 * 0.1 = 500$	6,500	-	6,500
4	$5,000 * 0.1 = 500$	7,000	-	7,000
5	$5,000 * 0.1 = 500$	7,500	7,500	
Total			7,500	

2) Compound interest, pay all at end

Plan 2

End of Year	Interest for Year	Owed at End of Year	End of Year Payment	Owed after Paid
0				$5,000
1	$5,000 * 0.1 = 500$	5,500	-	5,500
2	$5,500 * 0.1 = 550$	6,050	-	6,050
3	$6,050 * 0.1 = 605$	6,655	-	6,655
4	$6,655 * 0.1 = 665,5$	7320,5	-	7320,5
5	$7320,5 * 0.1 = 732.05$	8052.55	8052.55	
Total			8052.55	

3) Simple interest paid annually, principal paid at end

Plan 3

End of Year	Interest for Year	Owed at End of Year	End of Year Payment	Owed after Payment
0				$5,000
1	$5,000 * 0.1 = 500$	5,500	500	5,000
2	$5,000 * 0.1 = 500$	5,500	500	5,000
3	$5,000 * 0.1 = 500$	5,500	500	5,000
4	$5,000 * 0.1 = 500$	5,500	500	5,000
5	$5,000 * 0.1 = 500$	5,500	5,500	
Total			7,500	

4) Compound interest and portion of principal ($1,000) repaid annually

Plan 4

End of Year	Interest for Year	Owed at End of Year	End of Year Payment	Owed after Payment
0				$5,000
1	$5,000 * 0.1 = 500$	5,500	1,500	4,000
2	$4,000 * 0.1 = 400$	4,400	1,400	3,000
3	$3,000 * 0.1 = 300$	3,300	1,300	2,000
4	$2,000 * 0.1 = 200$	2,200	1,200	1,000
5	$1,000 * 0.1 = 100$	1,100	1,100	
Total			6,500	

Terminology and Symbols

- *P*: value or amount of money at present or time 0.
  - Present Worth (PW)
  - Present Value (PV)
  - Net Present Value (NPV)
- *F*: value or amount of money at some future time.
  - Future Worth (FW)
  - Future Value (FV)
- *A*: series of consecutive, equal, end-of-period amounts of money.
- *n*: number of interest periods
- *i*: interest rate or rate of return per time period
  - compound unless otherwise stated
  - expressed as percent per interest period
- *t*: time stated in periods

EXAMPLE

- Today, Julie borrowed $5000 to purchase furniture for her new house. She can repay the loan in either of the two ways described below. Determine the engineering economy symbols and their value for each option.

  (*a*) Five equal annual installments with interest determined at 5% per year.

  (*b*) One payment 3 years from now with interest determined at 7% per year.

Solution

- (*a*) The repayment schedule requires an equivalent annual amount *A*, which is unknown.

  P = \$5000 i = 5% per year n = 5 years A = ?

- (*b*) Repayment requires a single future amount *F*, which is unknown.

  P = \$5000 i = 7% per year n = 3 years F = ?

EXAMPLE

- You plan to make a lump-sum deposit of $5000 now into an investment account that pays 6% per year, and you plan to withdraw an equal end-of-year amount of $1000 for 5 years, starting next year. At the end of the sixth year, you plan to close your account by withdrawing the remaining money. Define the engineering economy symbols involved.

Solution

- All five symbols are present, but the future value in year 6 is the unknown.
  - P = $5000
  - A = $1000 per year for 5 years
  - F = ? at end of year 6
  - i = 6% per year
  - n = 5 years for the A series and 6 for the F value

EXAMPLE

- Last year Jane's grandmother offered to put enough money into a savings account to generate $5000 in interest this year to help pay Jane's expenses at college.

  (*a*) Identify the symbols, and

  (*b*) calculate the amount that had to be deposited exactly 1 year ago to earn $5000 in interest now, if the rate of return is 6% per year.

Solution

- (*a*) Symbols *P* (last year is −1) and *F* (this year) are needed.
  - *P* = ?
  - *i* = 6% per year
  - *n* = 1 year
  - *F* = *P* + interest = ? + $5000

(*b*) Let F = total amount now and P = original amount. We know that $F - P = \$5000$ is accrued interest.

Now we can determine P.

$$F = P + Pi$$

- The \$5000 interest can be expressed as

$$\text{Interest} = F - P = (P + Pi) - P$$

$$= Pi$$

$$\$5000 = P(0.06)$$

$$P = \frac{\$5000}{0.06} = \$83{,}333.33$$